Beam Velocity and Density Influences on Ion-Beam Pulses Moving in Magnetized Plasmas

Xiao-Ying Zhao, Hong-Peng Xu, Yong-Tao Zhao, Xin Qi, and Lei Yang

***Abstract*— The wakefield and stopping power of an ion-beam pulse moving in magnetized plasmas are investigated using particle-in-cell simulations. The effects of beam velocity and density on the wakefield and stopping power are discussed. Besides the longitudinal inverted V-shaped wakes, strong whistler waves are observed when low-density and low-velocity pulses move in plasmas in the presence of a magnetic field. The corresponding stopping powers are enhanced because of drag from these whistler waves. As the beam velocities increase, the whistler waves disappear, and only inverted V-shaped wakes are observed. The corresponding stopping powers are reduced compared with those in isotropic plasmas. When high-density pulses are transported in the magnetized plasmas, the whistler waves are greatly inhibited for low-velocity pulses and disappear for high-velocity pulses. In addition, the magnetic field reduces the stopping powers for all high-density cases.**

***Index Terms*— Magnetized plasmas, particle-in-cell (PIC) simulation, stopping power, wakefield.**

I. Introduction

UNDERSTANDING the interactions of charged particles with magnetized plasmas has been an interesting theme among researchers for several decades. The stopping power of ion beams and plasma waves excited by the charged particles are important for fundamental physics [1] and many applications such as electron cooling of ion beams [2], [3], inertial confinement fusion driven by ion beams [4], [5], and neutral beam injection in magnetically confined fusion plasmas [6], [7].

When charged particles are injected into the plasmas, the wakefield exhibits characteristic structures arising from the response of plasma electrons to the disturbance from the charged particles [8]–[13]. The structure of the wakefield is important in a plasma-wakefield accelerator [14], [15], where the wakefield induced by the ion beams can be used to accelerate the charged particles. For isotropic plasmas, Kaganovich *et al.* [11] developed the analytical electron fluid model to describe the plasma response to a propagating ion beam. They showed that the ion beam is neutralized by plasmas and forms a V-shaped cone structure in the wakefield region. However, when ion beams move in a magnetized plasma, wakes are more complicated. The wakefield is highly asymmetrical when a strong magnetic field is imposed perpendicular to the trajectory of a test charged particle [12]. For ion beams moving parallel to the external magnetic field, the wakes lose their typical V-shaped cone structure and exhibit an inverted V-shaped structure [9]. Dorf *et al.* [16] performed a linear theoretical analysis demonstrating that electromagnetic wave-field perturbations propagating oblique to the beam axis are excited when the magnetic field strength satisfies $\omega_{ce} > 2\beta_b\omega_{\rm pe}$ and the ion beam pulse is sufficiently long $l_b \gg V_b/\omega_{\rm pe}$. Here, ω_{ce} and $\omega_{\rm pe}$ are the electron cyclotron frequency and electron plasma frequency, respectively. $\beta_b = V_b/c$ is the ion beam velocity normalized to the speed of light, and l_b is the length of the ion beam pulse.

Another important quantity with regard to ions moving in magnetized plasmas is the stopping power, which is defined as the energy change per unit path length dE/ds. When pulses are transported in magnetized plasmas, the movement of the plasma electrons is restricted in the magnetic field direction. The influence of the magnetic field on the nonlinear stopping power in plasmas has been widely investigated for many years [18]–[25]. For low ion velocity ($V_{b0} < 10V_{\rm th}$), an enhancement of the stopping power for ions moving transverse to the magnetic field was found in an electron plasma [24], where $V_{\rm th} = (k_B T_e/m_e)^{1/2}$ is the thermal velocity of plasma electrons. Also, if the ion moves parallel to the magnetic field, the energy loss diminishes [25]. Moreover, collective effects [8], [26], [27] play an important role for high-density beam pulses transport in magnetized plasmas. Indeed, strengthening the magnetic field reduces the stopping power for low- and high-density pulses. In regions of moderate beam density, the stopping power increases in a weak magnetic field, but decreases in a strong magnetic field [9].

In our study, 2-D3V particle-in-cell (PIC) simulations were performed to investigate the process of a hydrogen ion-beam pulse moving longitudinal through hydrogen magnetized two-component plasmas. The effects of ion-beam pulse with different velocities and densities on the wakefield and stopping power were analyzed. All simulations were performed using the code VORPAL [28]. This paper is organized as follows.

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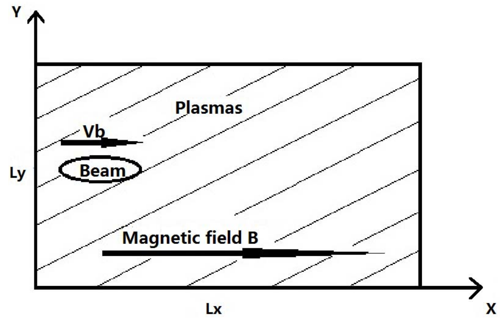

Fig. 1. 2-D plasma slab model.

In Section II, the PIC simulation methods used in this paper are briefly described. In Section III, the effects of beam velocity and density on the wake field and stopping power are presented and discussed. A summary is given in Section IV.

II. Particle-in-Cell Simulation Methods

We regard the plasma as an ensemble of charged particles, configured as a 2-D plasma slab (Fig. 1). The external magnetic field B applied in the plasmas is uniform and directed along the x-axis. The simulation region ranges from $x = 0$ to $x = L_x$ and $y = 0$ to $y = L_y$. Initially, the plasmas (including plasma electrons and ions) are placed in the simulation box; the ion-beam pulse enters from the left side and moves in the x-direction with an initial velocity V_b. The simulation uses a 2-D3V electromagnetic PIC code. All the charged particles are considered to move in the xy plane.

The equations of motion for all particles involved in the simulations (plasma electrons, ions, and the injected beam ions) are

$$\frac{d\mathbf{r}_j^\alpha}{dt} = \mathbf{v}_j^\alpha \tag{1}$$

$$\frac{d\left(\gamma_j^\alpha \mathbf{v}_j^\alpha\right)}{dt} = \frac{q_\alpha}{m_\alpha}\left(\mathbf{E} + \mathbf{v}_j^\alpha \times \mathbf{B}\right), \quad j = 1, 2, \ldots, N_\alpha. \tag{2}$$

Here, $\mathbf{r}_j^\alpha$, $\mathbf{v}_j^\alpha$, q_α, γ_j^α, m_α, and N_α are the position, velocity, charge, Lorentz factor, mass, and total number of plasma electrons ($\alpha = e$), ions ($\alpha = i$), and injected beam ions ($\alpha = b$), respectively. The electric and magnetic fields are updated using the Faraday equation and the Ampere–Maxwell equation

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} \tag{3}$$

$$\frac{\partial \mathbf{E}}{\partial t} = c^2 \nabla \times \mathbf{B} - \frac{\mathbf{J}}{\varepsilon_0} \tag{4}$$

where the particle source is

$$\mathbf{J} = \sum_j q_\alpha \mathbf{v}_j^\alpha \delta(x - x_j). \tag{5}$$

All simulations were performed using VORPAL software. The simulation box was meshed with 800 grid lines along the x-axis and 512 grid lines along the y-axis. The space and time steps are set to $dx = dy = \lambda_e = 5.25 \times 10^{-5}$ m and $dt = 9.1 \times 10^{-14}$ s, respectively, which satisfy the Courant–Friedrichs–Lewy limit, where $\lambda_e = (\varepsilon_0 K_B T_{e0}/N_{e0} e^2)^{1/2}$ is the Debye length of plasma electrons. Fifty superparticles are placed in each cell in simulations. A periodic boundary condition in the y-direction and an open boundary condition in the x-direction are adopted. The plasmas are composed of electrons and H ions. A uniform plasma density $N_{e0} = N_{i0} = 2.0 \times 10^{17}$ m^{-3} and an initial plasma temperature $T_{e0} = T_{i0} = 10$ eV are used for electrons and ions, respectively. Composed of protons, the ion beam has a profile density obeying the Gaussian distribution: $N_b(x, y) = \rho_{b0} \exp(-((x - x_0)^2/(L_b/2)^2)) \exp(-((y - y_0)^2/R_b^2))$, where the length $L_b = 40 \times \lambda_e$ and radius $R_b = 2 \times \lambda_e$. All particles are considered to be charged rods moving in the xy plane, and the simulation tracks the following coordinates for each particle: x, y, V_x, V_y, and V_z, which goes partway to a 3-D model. At each time step, the energy loss ΔE and the travel path Δs of each injected ion are recorded. The stopping power per ion is calculated by averaging $\Delta E/\Delta s$ over the entire simulation and over all the particles.

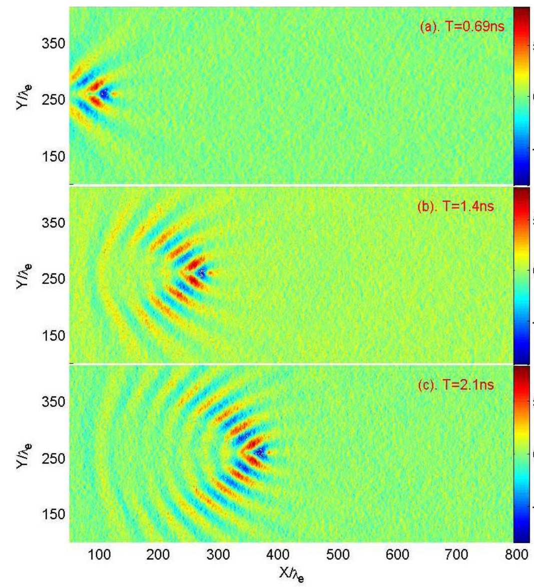

Fig. 2. Time evolution of an ion-beam pulse injected into a magnetized plasma with magnetic field $B = 1.0$ T, injection velocity $V_{b0} = 0.025c$, and beam density $\rho_{b0} = 1.0 \times 10^{17}$ m^{-3}. The electric field E_{ind} ($\times 10^4$ V/m) induced by the pulse at three different points in time are displayed. (a) $T = 0.69$ ns. (b) $T = 1.4$ ns. (c) $T = 2.1$ ns.

III. Simulation Results and Discussion

The influence of beam velocity on the wakefield and stopping power when beam density is smaller than electron density $\rho_{b0} = 1.0 \times 10^{17}$ m^{-3} (the linear case) is investigated first. Fig. 2 shows the evolution of the electric wakefield induced by the ion-beam pulse moving in a magnetic plasma when the beam velocity is $V_b = 0.025c$ (satisfying $L_b \gg V_b/\omega_{\text{pe}}$). At the very beginning [Fig. 2(a)], the perturbations of the whistler wave field propagating oblique to the beam axis is observed, which is consistent with the linear theoretical analysis prediction in [16]. When $T = 1.4$ ns [Fig. 2(b)],

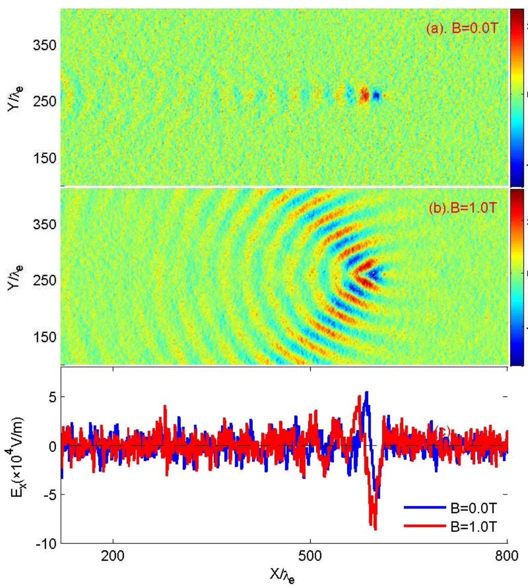

Fig. 3. Contour plot of the electric field E_{ind} ($\times 10^4$ V/m) in the wakefield region induced by the charged particles moving in the plasmas at the beam velocity $V_{b0} = 0.025c$, beam density $\rho_{b0} = 1.0 \times 10^{17}$ m^{-3}, and time $t = 3.6$ ns for (a) $B = 0.0$ T and (b) $B = 1.0$ T. (c) Corresponding longitudinal electric field along the trajectory of the pulse. The blue line is $B = 0.0$ T, and the red line is $B = 1.0$ T.

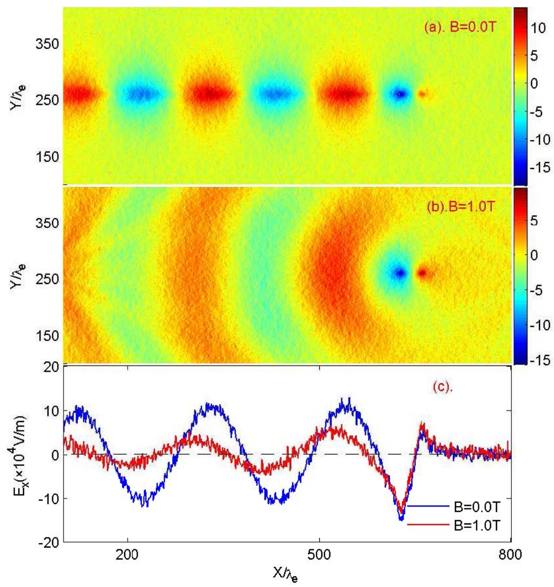

Fig. 4. Contour plot of the electric field E_{ind} ($\times 10^4$ V/m) in the wakefield region induced by the charged particles moving in the plasmas at the beam velocity $V_{b0} = 0.15c$, beam density $\rho_{b0} = 1.0 \times 10^{17}$ m^{-3}, and time $t = 0.68$ ns for (a) $B = 0.0$ T and (b) $B = 1.0$ T. (c) Corresponding longitudinal electric field along the trajectory of the pulse. The blue line is $B = 0.0$ T, and the red line is $B = 1.0$ T.

the wakes exhibit whistler waves, but no obvious structure is seen behind the trajectory. In Fig. 2(c), when $T = 2.1$ ns, apart from the whistler waves, inverted V-shaped cone structures are observed. To further investigate the influence of the magnetic field on the wakefield, a comparison (Fig. 3) was made of the electric wakefield induced by the ion-beam pulse moving in isotropic plasmas [Fig. 3(a)] and magnetized plasmas [Fig. 3(b)]. At $B = 0.0$ T [Fig. 3(a)], the wakes exhibit a fluctuation behind the pulse. However, at $B = 1.0$ T [Fig. 3(b)], the wakes show both an inverted V-shaped cone and a whistler wave structure. Moreover, the amplitude of the whistler waves is larger than that of the inverted V-shaped fluctuation. The corresponding longitudinal electric field along the trajectory of the pulse for isotropic and magnetic cases is given in Fig. 3(c). For low-density and low-velocity beam pulses, the length of ion-beam pulse is much greater than the wavelength of electron plasma wave excitations, $L_b \gg V_b/\omega_{\text{pe}}$. Therefore, the electrostatic electron plasma wave excitations are significantly suppressed and there is no obvious wave just behind the ion-beam pulse for both cases.

For a higher velocity, $V_{b0} = 0.15c$ ($L_b \approx V_b/\omega_{\text{pe}}$), the results are given for the wakefield in isotopic plasmas [Fig. 4(a)] and in magnetized plasmas [Fig. 4(b)] where the inverted V-shaped cone structures are observed when $B = 1.0$ T. Compared with the structures in Fig. 3(b), the whistler waves disappear in the wake region. This is because for beam pulses with high velocity, the electrostatic electron plasma wave excitations play a dominant role. Fig. 4(c) shows the longitudinal electric field along the trajectory for both cases. In the presence of the magnetic field, the wakefield is clearly inhibited. This is because plasma electrons are restricted in the direction perpendicular to the magnetic field.

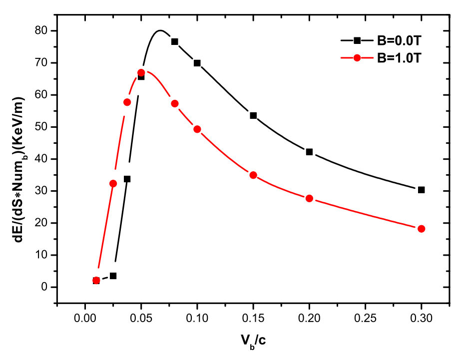

Fig. 5. Stopping power per ion as a function of beam velocity when beam density $\rho_{b0} = 1.0 \times 10^{17}$ m^{-3} with different magnetic fields. The black square corresponds to $B = 0.0$ T and the red circle to $B = 1.0$ T. The line is a linear fit of the simulation data.

Fig. 5 shows the dependence of the stopping power per pulse ion on the beam velocity with different magnetic fields, $B = 0.0$ T and $B = 1.0$ T. Note that for low-velocity beams ($V_{b0} < 0.05c$), the presence of magnetic field enhances the stopping power. This is because in the presence of the magnetic field, whistler waves are excited by the ion-beam pulse [Fig. 3(b)]. These wakes draw back the beam and enhance the stopping power. As velocity increases ($V_{b0} > 0.05c$), the stopping power diminishes in the presence of the magnetic field.

The above discussion mainly concerns conditions when the density of the pulses is smaller than the electron density in plasmas. When the density of beam pulses is much higher than the density of the electrons, the collective effects and screening electrons play important roles. Fig. 6 shows the electric wakefield induced by the ion-beam pulse for isotropic

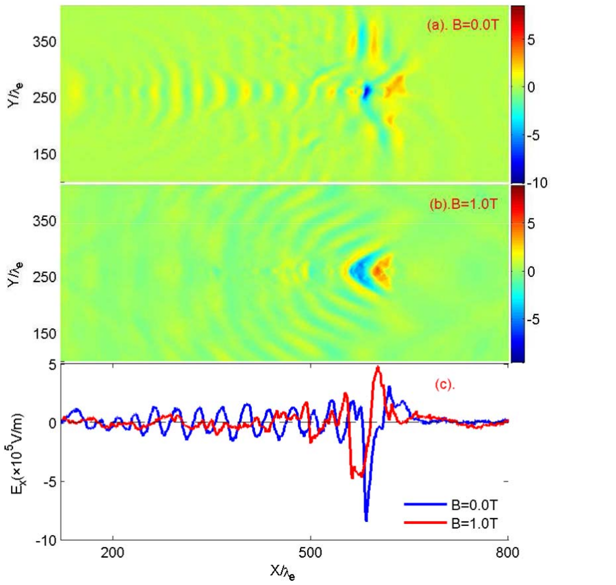

Fig. 6. Contour plot of the electric field E_{ind} ($\times 10^5$ V/m) in the wakefield region induced by the charged particles moving in the plasmas at the beam velocity $V_{b0} = 0.025c$, beam density $\rho_{b0} = 2.0 \times 10^{18}$ m^{-3}, and time $t = 3.6$ ns for (a) $B = 0.0$ T and (b) $B = 1.0$ T. (c) Corresponding longitudinal electric field along the trajectory of the pulse. The blue line is $B = 0.0$ T, and the red line is $B = 1.0$ T.

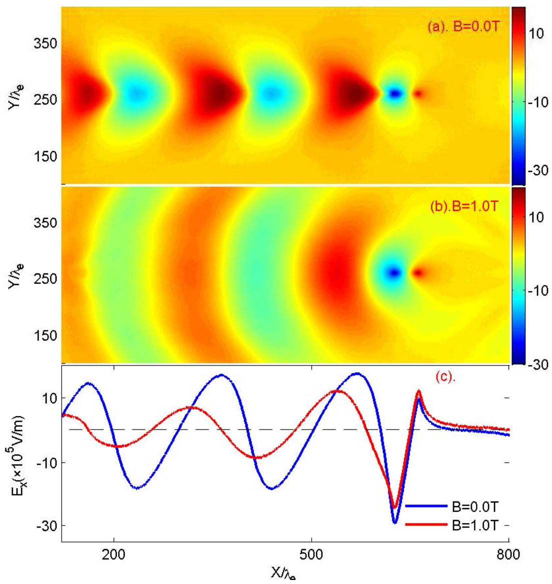

Fig. 7. Contour plot of the electric field E_{ind} ($\times 10^5$ V/m) in the wakefield region induced by the charged particles moving in the plasmas at the beam velocity $V_{b0} = 0.15c$, beam density $\rho_{b0} = 2.0 \times 10^{18}$ m^{-3}, and time $t = 0.68$ ns for (a) $B = 0.0$ T and (b) $B = 1.0$ T. (c) Corresponding longitudinal electric field along the trajectory of the pulse. The blue line is $B = 0.0$ T, and the red line is $B = 1.0$ T.

plasma and magnetic plasma when $\rho_{b0} = 2.0 \times 10^{18}$ m^{-3} and $V_{b0} = 0.025c$. At $B = 0.0$ T [Fig. 6(a)], the wakes exhibit typical V-shaped structures behind the pulse; at $B = 1.0$ T [Fig. 6(b)], whistler waves are observed. However, in comparison with Fig. 3(b), the amplitude of the whistler waves is obviously reduced. This is because for high-density pulses propagating in plasmas, nonlinear perturbations caused by collective effects weaken the whistler wave excitations. The longitudinal electric field along the trajectory is shown in Fig. 6(c). As discussed above, the wakefield is inhibited in the presence of the magnetic field. From the results for higher velocity, when $V_{b0} = 0.15c$ (Fig. 7), the wakes exhibit very apparent V-shaped cone structures when $B = 0.0$ T [Fig. 7(a)]. The inverted V-shaped cone structures appear when $B = 1.0$ T [Fig. 7(b)]. Moreover, as the velocity increases, the whistler waves disappear. The longitudinal electric fields along the trajectory are given in Fig. 7(c). In the presence of the magnetic field, the amplitude of the wakefield clearly diminishes. This is because, in a magnetic field, more electrons are restricted in the transverse direction and cannot actively respond to the attraction of the pulses. Moreover, the magnetic field reduces the screening of the ion-beam pulses. As electron screening weakens, the collective effect of the ion-beam pulse becomes significant, which enhances electron trapping and inhibits the fluctuation of the wakefield.

In Fig. 8, we give the stopping power per pulse ion when the beam density $\rho_{b0} = 2.0 \times 10^{18}$ m^{-3}. The results for pulses in isotropic and magnetized plasmas are shown for comparison. One can see that for high-density pulses, the magnetic field reduces the stopping power for all beam velocities. This behavior occurs because the magnetic field reduces the screening of the ion-beam pulses. As the screening effect of the electrons

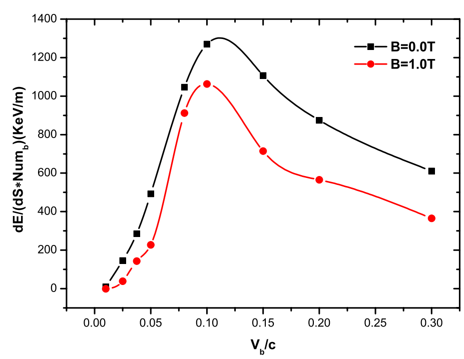

Fig. 8. Stopping power per ion as a function of beam velocity when beam density $\rho_{b0} = 2.0 \times 10^{18}$ m^{-3} with different magnetic fields. The black square corresponds to $B = 0.0$ T and the red circle to $B = 1.0$ T. The line is a linear fit of the simulation data.

weakens, the collective effects of the ion-beam pulses become more significant. This enhances the electron trapping processes and, consequently, decreases the stopping power. Moreover, for a low-velocity pulse when $V_{b0} = 0.025c$, the whistler wave excitations are inhibited, which leads to a reduction in stopping power.

IV. Conclusion

In this paper, we use 2-D3V PIC simulations to study the wakefield and stopping power of an ion-beam pulse moving in magnetized plasmas. Beams moving in plasmas in the absence of a magnetic field were also studied for comparison. The influence of beam velocity and density on the wake and stopping power is investigated.

For beam pulses with low beam density (the beam densities are smaller than the plasma electron density), the wakes induced by low-velocity pulses exhibit inverted V-shaped structures. Moreover, strong whistler waves were observed on both sides of the beam trajectory. The corresponding stopping power is enhanced because of drag from these whistler waves. As beam velocities increase, the whistler waves disappear, and only inverted V-shaped wakes are observed. The corresponding stopping powers decrease compared with those in isotropic plasmas.

For beam pulses of high density (much greater than the plasma electron densities), collective effects dominate. The whistler wave excitations are inhibited for low-density pulses. As the velocity increases, whistler waves disappear, and only inverted V-shaped wakes are observed. The presence of the magnetic field enhances the collective effects as well as electron trapping, and consequently reduces the stopping power for all beam velocities.

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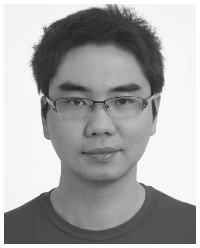

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